

Dwip Dalal

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Education

- **University of Illinois Urbana-Champaign (UIUC)** 2024 – 2028 (Expected)
Ph.D. Candidate, Electrical and Computer Engineering; Advisor: [Prof. Svetlana Lazebnik](#)
GPA: 4.0/4.0
- **Indian Institute of Technology Gandhinagar (IIT-GN)** 2020 – 2024
B.Tech in Mechanical Engineering with Minor in Computer Science
[Institute Gold Medalist](#); GPA: 9.1/10; Minor GPA: 10/10

Research Interests

World Action Models (WAMs), Vision-Language-Action Models (VLAs), Robot Learning, Multimodal Representation Learning, VLMs, Reasoning, Test-Time Scaling

Research Internships

- **Applied Scientist Intern @ Amazon** (May 2026 – Aug 2026)
Mentor: [Dr. Sowndarya Sundar](#), [Dr. Keval Doshi](#), [Dr. Jim Thomas](#) *Location: Sunnyvale*
Improved MLLMs' long-video understanding through efficient and targeted evidence acquisition. [2]
- **ML Research Intern @ Microsoft Research** (May 2025 – Mar 2026)
Mentor: [Dr. Nebojsa Jojic](#) (Senior Principal Researcher) *Location: Redmond*
Built CityNav, an environment for evaluating MLLMs on long-horizon outdoor navigation, and VoP, a method that elicits latent knowledge from MLLMs, achieving SOTA performance on instruction-free navigation [4].
- **ML Research Intern @ University of British Columbia (MITACS GRI)** (May 2023 – Aug 2023)
Mentor: [Prof. Yankai Cao](#) *Location: Vancouver*
Built a neural-to-video generation framework by learning the joint distribution between video and spike-train (brain activity) that reconstructs mouse behavior directly.
- **NLP Research Intern @ University of South Carolina** (Aug 2022 – Feb 2024)
Mentor: [Prof. Amit Sheth](#) (Director of AI institute @ UoSC), [Prof. Amitava Das](#), [Dr. Aman Chadha](#)
Worked on multimodal fact verification for language models, [8, 9]; multitask learning [15]. Also, launched [DeHate](#), a challenge focused on the automatic blurring of image segments (De-Factify 3.0 @AAAI'24).

Academic Research Positions

- **Research Assistant, University of Illinois Urbana-Champaign** (Aug 24 – Present)
Mentor: [Prof. Svetlana Lazebnik](#), [Prof. Unnat Jain](#), [Prof. Heng Ji](#)
Currently working on improving the out-of-distribution (OOD) generalization of World Action Models. Improved OOD generalization in VLAs for robotic Manipulation [1], visual grounding in VLMs [3]; improved compositional reasoning of VLMs[5]; and authored a survey on agentic systems [10].
- **Research Assistant, IIT Gandhinagar** (Nov 21 – May 24)
Mentor: [Prof. Shanmuganathan Raman](#), [Prof. Anirban Dasgupta](#), [Prof. Mayank Singh](#), [Prof. Harish PM](#)
Worked on diffusion model for image-to-image tasks [7], cross-modal representation learning [6], normalizing flow for constrained geometry, density estimation and path-planning [11, 12], and topic modeling [13].

Awards & Achievements

- Awarded the *Dan Vivoli Endowed Fellowship* @ UIUC 2026–2027 for outstanding academic performance.
- Awarded the *Qualcomm Graduate Award* @ UIUC 2025–2026.
- Awarded the *Dilip and Sandhya Sarwate Graduate Fellowship* @ UIUC 2024–2025, awarded to outstanding incoming graduate students.
- Awarded the scholarship from Thinking Machines to support my research on VLAs.
- Awarded the scholarship from Modal to support my research on VLAs.
- Awarded the [Institute Gold Medal](#) @ IIT Gandhinagar, for the highest CGPA in the discipline.
- Selected for Y Combinator's AI Startup School 2026.

- 1st position out of 200 participants in a ML hackathon organized by SmartSense Consulting Solutions Pvt. Ltd., earning a job offer for the role of ML Engineer II (declined).
- Institute nomination for Pre-Doctoral Research Assistant Program at Microsoft Research, India.
- 1st rank in [Undergraduate Research Showcase '23](#) @ IITGN for work on AI-enabled on-device drones.
- 10/10 GPA for two consecutive semesters (6th & 7th), each comprising rigorous 26-credit course work.
- Awarded *Shastri Indo-Canadian Institute Scholarship* for my research at the University of British Columbia.
- 1st rank in [Hackrushi '22](#), the flagship annual machine learning hackathon of IIT Gandhinagar.
- 1st rank in IITGN Summer Project 2021 for building the most efficient NLP classification model.
- 1st rank in the competitive programming contest @ IIT-GN, participated by B.Tech, MTech, and Ph.D.

Publications

Under Review

- Generalizable VLA Finetuning via Representation Anchoring and Language-Action Alignment
Dwip Dalal, S. Patel, C. Jain, J. Kim, U. Mishra, A. Baratian, H. Ha, Heng Ji, Svetlana Lazebnik, Unnat Jain
Under review [\(Paper\)](#) [\(Project Page\)](#)
- Repairable Uncertainty: Selective Temporal Reacquisition for Long-Video Understanding with MLLMs
Dwip Dalal, K. Doshi, A. Kumar, W. Wang, S. Sundar, N. Vedapunt, J. Thomas, K. Fu, Svetlana Lazebnik
Under review [\(Paper\)](#) [\(Project Page\)](#)

Conference Papers

- Constructive Distortion: Improving MLLMs with Attention-Aware Image Warping
Dwip Dalal, G. Vashishtha, U. Mishra, J. Kim, M. Kanda, H. Ha, Svetlana Lazebnik, Heng Ji, Unnat Jain
ICLR 2026 [\(Paper\)](#) [\(Project Page\)](#)
- City Navigation in the Wild: Exploring Emergent Navigation from Web-Scale Knowledge in MLLMs
Dwip Dalal*, Utkarsh Mishra*, Narendra Ahuja, Nebojsa Jojic
EACL 2026 [\(Paper\)](#) [\(Project Page\)](#)
- Compositional Reasoning via Joint Image and Language Decomposition
Dwip Dalal*, Madhav Kanda*, Zhenhailong Wang, Heng Ji, Unnat Jain
EACL 2026 [\(Paper\)](#)
- Learning Robust Deep Visual Representations from EEG Brain Recordings
Prajwal Singh, **Dwip Dalal**, Gautam Vashishtha, Shanmuganathan Raman, Krishna Prasad Miyapuram
WACV 2024 (*Featured in WACV Daily and Best of WACV 2024*) [\(Paper\)](#)
- Single Image LDR to HDR Conversion using Conditional Diffusion
Dwip Dalal, Gautam Vashishtha, Prajwal Singh, Shanmuganathan Raman
ICIP 2023 [\(Paper\)](#)
- FACTIFY-5WQA: 5W Aspect-based Fact Verification through Question Answering
Anku Rani, SM Tonmoy, **Dwip Dalal**, Shreya Gautam, Megha C., Aman Chadha, Amit Sheth, Amitava Das
ACL 2023 [\(Paper\)](#) [\(Demo\)](#)
- A Benchmark for Multimodal Fact Verification with Explainability through 5W Question-Answering
M Chakraborty, K Pahwa, A Rani, S Chatterjee, **Dwip Dalal**, ... A Chadha, Amit Sheth, Amitava Das
EMNLP 2023 [\(Paper\)](#)

Technical Reports

- A Dataset-Centric Survey of LLM-Agents for Data Science
Dwip Dalal*, Chuxuan Hu*, Xiaona Zhou* [\(Paper\)](#)
- Flow Symmetrization for Parameterized Constrained Diffeomorphisms
Dwip Dalal*, Aalok Gangopadhyay*, Progyan Das*, Shanmuganathan Raman [\(Paper\)](#) [\(Project Page\)](#)

Workshop Papers

- ODESolvers are also Wayfinders: Neural ODEs for Multi-Agent Pathplanning
Dwip Dalal*, Progyan Das*, Anirban Dasgupta
NeurIPS 2023 Workshop (DLDE) [\(Paper\)](#)

13. MMT: A Multilingual and Multi-Topic Indian Social Media Dataset
Dwip Dalal, Vivek Srivastava, Mayank Singh
EACL 2023 Workshop (C3NLP) [\(Paper\)](#)
14. Stabilizing Swirling Pendulum using Actor-Critic Method
Dwip Dalal, Shubhankar Riswadkar, Harish J Palantheandaram-Madapusi
Indian Control Conference (ICC) 2023 [\(Paper\)](#)
15. SEPSIS: I can catch your lies - A new paradigm for Deception Detection
Anku Rani, Dwip Dalal, S. Gautam, P. Gupta, Vinija Jain, Aman Chadha, Amit Sheth, Amitava Das
ACL 2025 Workshop (SRW) [\(Paper\)](#)

Industrial Internships

- **ML Intern @ Physical Research Laboratory, ISRO** (Jan 23 – July 23)
Mentor: Prof. Dibyendu Chakrabarty (PI, ISRO Aditya-L1 Mission)
Implemented transformer-based pipeline with Bayesian inference on data from India's maiden solar probe, the Aditya-L1, to predict anomalous geomagnetic storms with uncertainty quantification.
- **ML Intern @ Tata Consultancy Services (TCS Research)** (Oct 22 – April 23)
Mentor: Dr. Manasi Patwardhan
Designed a transformer-based architecture for product copy generation, trained on the FACAD dataset. Fine-tuned BLOOM and ResNet models using the CLIP loss function to enable effective cross-modal learning.
- **AI Intern @ Defense Research and Development Organization (DRDO)** (Jan 22 – Aug 22)
Developed an RL framework using the actor-critic method in a PyBullet environment with simulated physics to learn drone swarm navigation.
- **SDE Intern @ Efcens Systems** (May 22 – Aug 22)
Developed an intelligent network threat detection platform to red-flag anomalies with Sanjeev Kumar, CTO of DELL, and Prasad Malempati, Sr. Manager of Accenture, San Francisco.

Relevant Courses

- **UIUC:** CS444 Deep Learning Computer Vision (A+); CS543 Computer Vision (A+); CS598 Topics in LLM Agents; CS546 Advanced Topics in NLP; CS 598 LLM Applications; ECE544 Pattern Recognition; ECE 598 Diffusion Model
- **IITGN:** Optimization for ML; Large Language Models; Digital Image Processing; Data Structures and Algorithms; Control Theory; Linear Algebra; Multivariable Calculus; Complex Analysis; Differential Equations; Probability; Numerical Methods

Open-Source Projects & Contributions

- **Merged Pull Request @ OpenVINO (Intel)** [\(Repository\)](#)
- **NeuralSight: Computer Vision Algorithm Library** [\(Repository\)](#)
- **JointGYM: Reinforcement Learning Algorithm Library** [\(Repository\)](#)
- **Jacobian Transpose Method for Controlling Robotic Manipulators** [\(Repository\)](#)

Social Initiatives, Leadership & Innovation

- **Instructor, SC 336: Mathematics of Machine Learning | IIT-GN | Fall, 2023** [\(certificate\)](#) [\(Page 25\)](#)
 - Instructed IIT Gandhinagar's first student-led, fully accredited short course, with class size of 170 students.
- **Founder & President of Society for Machine Learning, IIT-GN**
- **Reviewer:** CVPR'26, ICLR'26, EMNLP'25, NeurIPS'25, ICCV'25, ACL'25, AISTATS'25, ICML'25, NeurIPS'24
- **Teaching Assistant, UIUC:** CS 444: Deep Learning for CV (Fall'25), CS449: AI (Spring'25)
- **Teaching Assistant, IIT-GN:** ES413 Deep Learning (Spring'24), CS328 Data Science (Spring'23)
- **Core Member of Robotics Club, IIT-GN**
- **Represented IIT-GN in the Inter-IIT Tech Meet in 2021, 2022, 2023 (Led team of 10).**